

Solution:

- (a) Let $z = |x - y|$ for distinct x, y . A collision occurs if and only if p divides z , i.e.,

$$h_p(x) = h_p(y) \iff p \mid z. \quad (1)$$

The number of primes in $[m/2, m]$ is given by the Prime Number Theorem as

$$\Theta(m / \log m). \quad (2)$$

If z has k prime factors in $[m/2, m]$, then

$$z \geq (m/2)^k. \quad (3)$$

Since $z \leq U$, we obtain

$$k \leq \log_{(m/2)} U. \quad (4)$$

Thus, z has at most $O(\log U)$ prime factors in $[m/2, m]$.

Since $U \geq m$, the probability of choosing a prime p that divides z is at most

$$\frac{O(\log U)}{\Theta(m / \log m)} = O\left(\frac{\log U}{m}\right). \quad (5)$$

- (b) Let $z = x - y$, the collision condition is:

$$h_A(x) = h_A(y) \iff A(x - y) = 0 \iff Az = 0. \quad (6)$$

Since $x \neq y$, we have $z \neq 0$, meaning there exists at least one bit i such that $z_i = 1$.

Let $a_1, a_2, \dots, a_w$ be the columns of A , and express:

$$Az = z_1 a_1 + z_2 a_2 + \dots + z_w a_w. \quad (7)$$

Since $z_i = 1$ for at least one i , rewrite:

$$Az = a_i + \sum_{j \neq i} z_j a_j. \quad (8)$$

The elements of $a_1, a_2, \dots, a_w$ are chosen independently and uniformly at random from $\{0, 1\}$, thus each element of $\sum_{j \neq i} z_j a_j$ is independent and uniform. So the probability that $Az = 0$ is simply the probability that each element of $a_i = 0 - \sum_{j \neq i} z_j a_j$, this occurs with probability $1/2^\ell$.

Thus, the probability of a collision is:

$$\Pr[Az = 0] = \frac{1}{2^\ell} = \frac{1}{m}. \quad (9)$$

(c) Let $z = x - y$,

$$(h_{A,b}(x) = s) \wedge (h_{A,b}(y) = t) \Rightarrow A(x - y) = s - t \Leftrightarrow Az = s - t \quad (10)$$

For fixed s and t , the probability that $Az = s - t$ is the probability that each element of $a_i = (s - t) - \sum_{j \neq i} z_j a_j$, this occurs with probability $1/2^\ell$:

$$\Pr_A[Az = s - t] = \frac{1}{m} \quad (11)$$

Given A s.t. $Az = s - t$,

$$(h_{A,b}(x) = s) \wedge (h_{A,b}(y) = t) \Leftrightarrow b = Ax - s \quad (12)$$

Since b is a random 0-1 vector, we have

$$\Pr_b[b = Ax - s | Az = s - t] = \frac{1}{m} \quad (13)$$

Combining 11 and 13, we obtain:

$$\begin{aligned} & \Pr_{A,b}[(h_{A,b}(x) = s) \wedge (h_{A,b}(y) = t)] \\ &= \Pr_A[Az = s - t] \Pr_b[b = Ax - s | Az = s - t] \\ &= \frac{1}{m^2} \end{aligned} \quad (14)$$